Newton’s Laws: A Short Learning Pack

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1. Newton’s First Law

An object remains at rest, or continues moving at constant velocity in a straight line, unless a net external force acts on it. This tendency to resist a change in motion is called inertia. Mass is a measure of an object’s inertia. A book resting on a table stays at rest because the forces on it are balanced: gravity pulls downward and the table’s normal force pushes upward.

2. Newton’s Second Law

Newton’s second law states that the net force on an object equals its mass multiplied by its acceleration. The relationship is commonly written as F = m × a. If the net force increases while mass remains constant, acceleration increases. If mass increases while net force remains constant, acceleration decreases. Force is measured in newtons, mass in kilograms, and acceleration in metres per second squared.

3. Newton’s Third Law

When one object exerts a force on a second object, the second object exerts a force of equal magnitude and opposite direction on the first. These action–reaction forces act on different objects, so they do not cancel each other on a single object. A swimmer pushes water backward, and the water pushes the swimmer forward.

4. Balanced and Unbalanced Forces

Forces are balanced when their vector sum is zero. An object then has no acceleration: it may remain at rest or continue at constant velocity. Forces are unbalanced when the net force is non-zero, causing acceleration in the direction of the net force. The net force is the vector sum of all forces, not simply the largest force.

5. Worked Example

A 2 kg cart experiences a net horizontal force of 6 N. Using F = m × a, acceleration equals F divided by m: a = 6 N / 2 kg = 3 m/s². If the same 6 N net force acts on a 4 kg cart, its acceleration is 1.5 m/s². Before applying the second law, first understand net force. Always identify the net force before applying the second law.

6. Common Misconceptions

A force is not required to keep an object moving at constant velocity; a net force is required to change velocity. Action–reaction forces do not cancel because they act on different objects. Mass and weight are not

identical: mass measures inertia, while weight is the gravitational force acting on an object. A diagram should label forces acting on the chosen object and show their directions.